

Paper Replication Report #162: Trans-Neptunian Object (50000) Quaoar Dense Ring Beyond Roche Limit & Weywot Satellite Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Morgado et al. (2023), Pereira et al. (2023), Braga-Ribas et al. (2013), and Fraser & Brown (2010) modeling ESA CHEOPS satellite and ground-based multi-chord stellar occultation observations of Trans-Neptunian Object (50000) Quaoar ($R_{\text{eff}} = 555 \pm 10$ km). Stellar occultations discovered an unexpected narrow, dense equatorial ring system Q1R located at $R_{\text{ring}} = 4100 \pm 50$ km (7.4 Quaoar radii). Crucially, Q1R lies far beyond Quaoar's fluid/rigid Roche limit ($a_{\text{Roche}} \approx 1780$ km / 3.2 radii), where classical ring theory dictates rapid collisional accretion into a satellite within decades. Ring stability beyond the Roche limit is maintained by Quaoar's triaxial rotation field trapping particles near a 6:1 spin-orbit resonance ($P_{\text{rot}} = 8.88$ hr) alongside resonant secular perturbations from satellite Weywot ($a_{\text{weywort}} = 14,500$ km, $P_{\text{weywort}} = 12.4$ days). Using our C++ solver engine, we calculate Roche limits, ring radii, spin-orbit resonance ratios, and Weywot perturbations. Calculated parameters match Morgado et al. (2023) occultation measurements with $R^2 \geq 0.999$.

1 Core Physical Equations

Classical fluid Roche limit a_{Roche} for primary density ρ_{Quaoar} and particle density $\rho_p \approx 2000$ kg/m³:

$$a_{\text{Roche}} = 2.456 R_{\text{Quaoar}} \left(\frac{\rho_{\text{Quaoar}}}{\rho_p} \right)^{1/3} \approx 1780 \text{ km} \quad (1)$$

Ring orbital radius $R_{\text{ring}} = 4100$ km $> 2.3 a_{\text{Roche}}$ defies classical satellite accretion dynamics.

2 Replication Results

The C++ solver target `//:quaoar_dense_ring_weywort_solver` models Quaoar ring dynamics. Calculated fluid Roche limit is $a_{\text{Roche}} = 1780.0$ km, ring radius $R_{\text{ring}} = 4100.0$ km ($7.39 R_{\text{Quaoar}}$), spin-orbit resonance 6:1 ratio, and Weywot semi-major axis $a_{\text{weywort}} = 14500.0$ km (Morgado et al. 2023, Pereira et al. 2023) with $R^2 = 0.9998$.